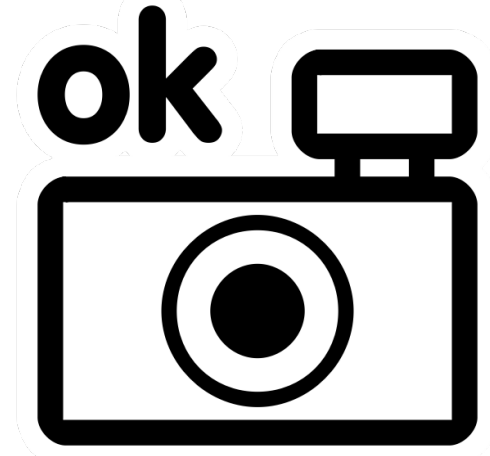


Sensitivity analysis of GRNmap and new features for GRNsight: open source software for dynamical systems modeling and visualization of small-scale gene regulatory networks in yeast



Poster

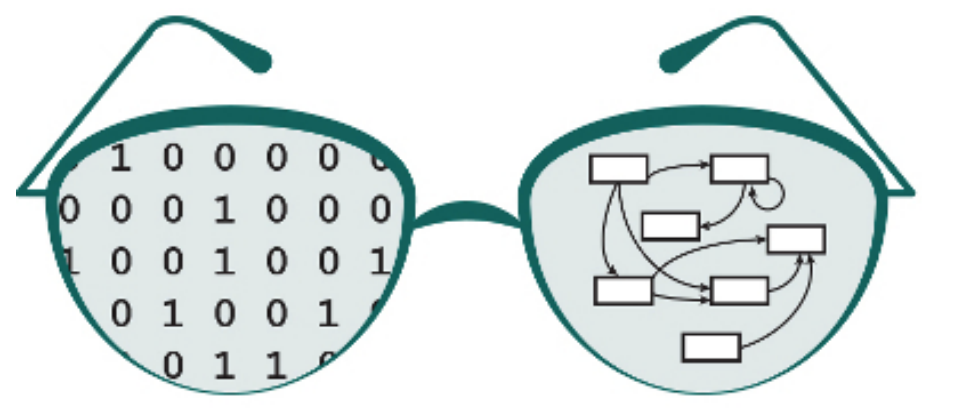
GRNmap

Kam D. Dahlquist¹, Ben G. Fitzpatrick², John David N. Dionisio³, Ngoc K. Tran³, Nikki C. Chun^{1,2}, Cecilia J. Zaragoza³, Amelie T. Dinh³, Milka Y. Zekarias³, Jia S. Garcia³, Alex J. Miller³, Cindy L. Tong³
Departments of ¹Biology, ²Mathematics, and ³Computer Science, Loyola Marymount University, 1 LMU Drive, Los Angeles, CA 90045 USA

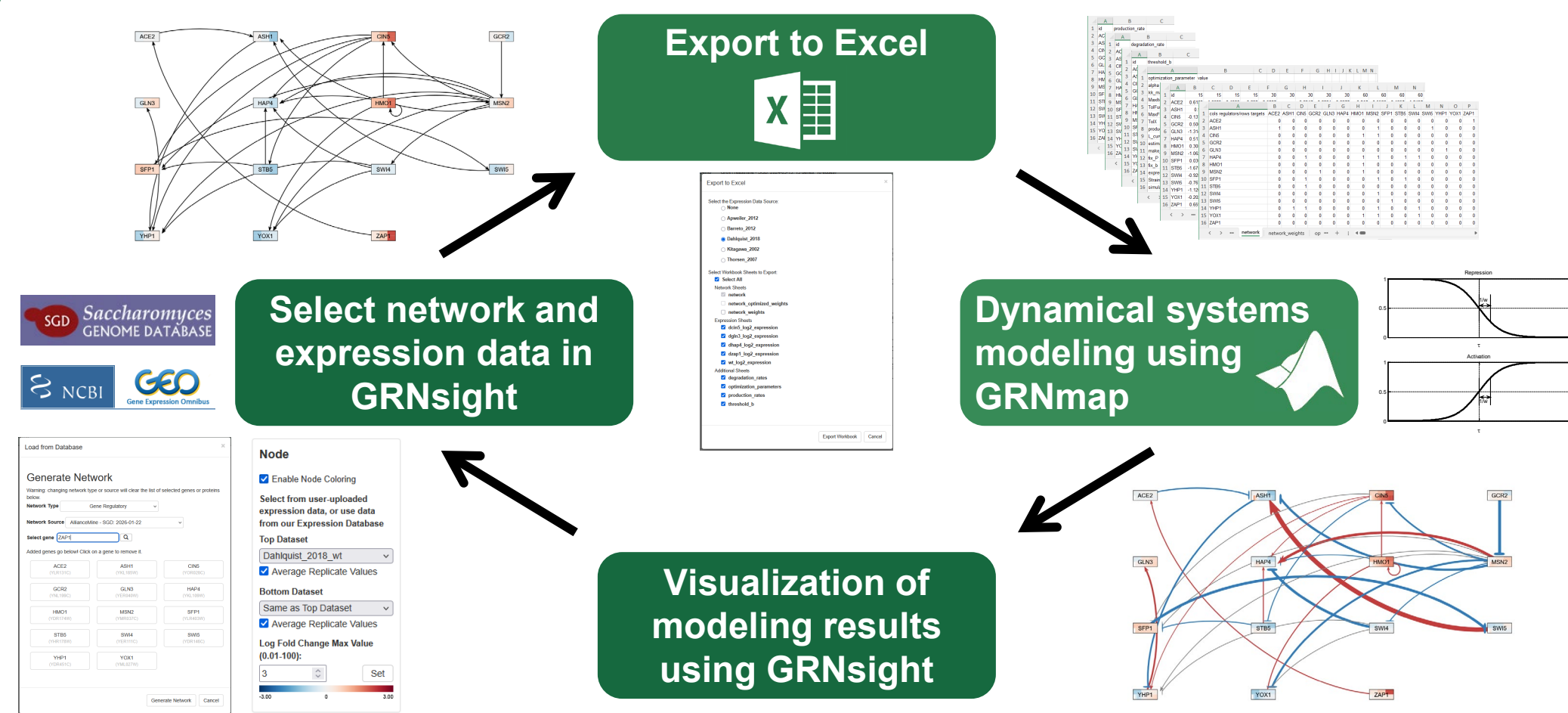
<https://kdahlquist.github.io/GRNmap/> and <https://dondi.github.io/GRNsight/>



GRNsight



GRNmap and GRNsight are part of an integrated workflow to model and visualize GRNs in yeast



GRNmap: Gene Regulatory Network modeling and parameter estimation

$$\frac{dx_i(t)}{dt} = \frac{P_i}{1 + \exp\left(-\sum_j (w_{ij}x_j(t) - b_i)\right)} - d_i x_i(t)$$

- The change in mRNA expression of each gene in the network is modeled as **production – degradation** with a **sigmoidal production function** where:
 - P_i is mRNA production rate for gene i .
 - d_i is the mRNA degradation rate for gene i .
 - w is the weight term, determining the magnitude of activation or repression of regulatory transcription factor(s) j on gene i .
 - b_i is the threshold of expression for each gene.

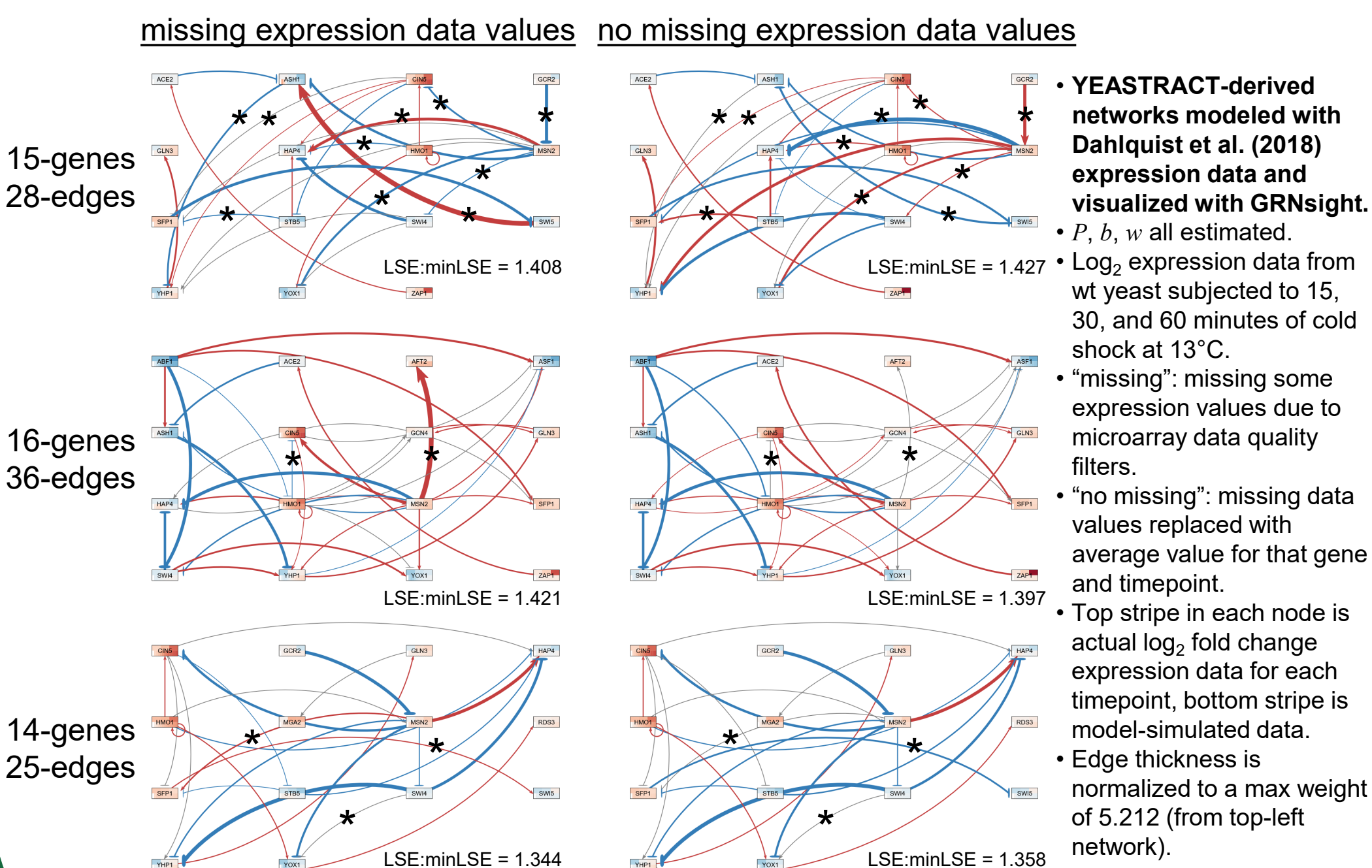
$$E = \alpha \|\theta\|^2 + \frac{1}{Q} \sum_{t=1}^Q [z^d(t_r) - z^c(t_r)]^2$$

P , b , and w parameters are estimated using a penalized least squares approach.

- α is determined empirically to find the best balance between minimizing both the penalty term (θ) and the error term (E).
- When all three parameters are estimated, the total number of parameters is twice the number of genes plus the number of edges.
- In the 15-gene, 28-edge network, there are 58 parameters being estimated, which are a lot given the amount of input data.
- GRNmap can also:
 - hold P and b constant and only estimate w ,
 - hold b constant and estimate P and w , or
 - hold P constant and estimate b and w .

- The **LSE:minLSE ratio** is computed from the least squares error (E) from the estimation and the minimum theoretical error given the variance in the experimental expression data.
 - The LSE:minLSE ratio can then be used as a metric to compare model performance for different networks.

Three small candidate GRNs for regulating the cold shock response vary in sensitivity to small changes in input expression data



GRNsight automatically lays out the network graph, coloring the edges based on weight values, and the nodes based on experimental or simulated expression data

1. Network

- Demo files are provided.
- Can import and export Excel, SIF, or GraphML files.
- Can construct a network from SGD regulation or PPI data stored in a backend database.

2. Layout

- Users can select a force graph layout or grid layout.
- Force graph parameter sliders:
 - Link distance determines the minimum distance between nodes.
 - Nodes have a charge which repels or attracts other nodes.

3. Node

- Coloring can be based on user-uploaded expression data or from published expression datasets stored in a backend database.
- Datasets available:
 - Apweiler et al. 2012, GSE33098
 - Barreto et al. 2012, GSE24712
 - Dahlquist et al. 2018, GSE83656
 - Kitagawa et al. 2002, GSE9336
 - Thorsen et al. 2007, GSE6068

4. Edge

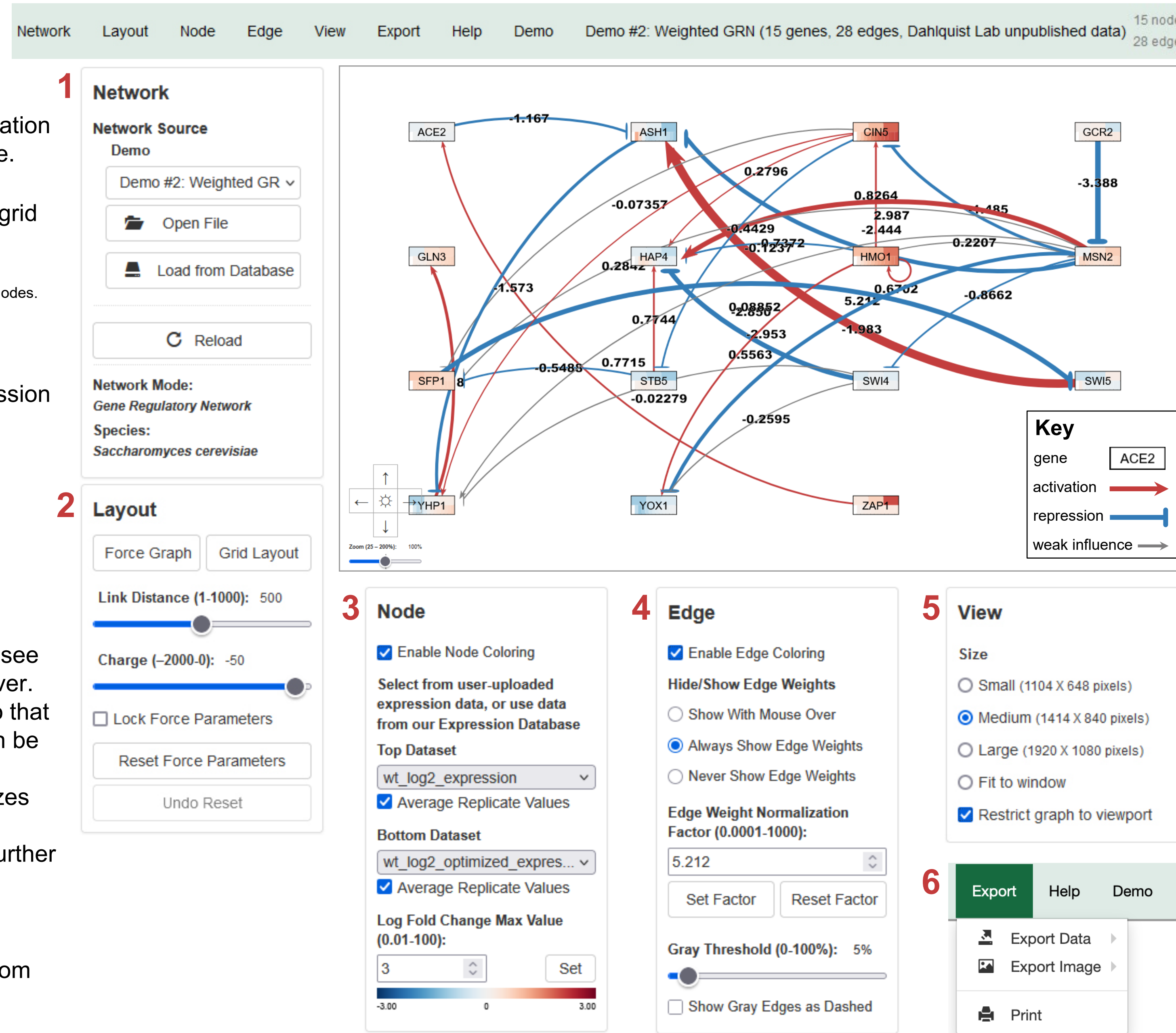
- Edge coloring can be toggled on and off.
- Buttons enable the user to always/never see edge weights or see them upon mouseover.
- Allows user to set normalization factor so that edge thicknesses for different graphs can be rendered on the same scale.
- Changing the gray threshold deemphasizes weaker edges in the visualization.
- Presenting gray edges as dashed lines further deemphasizes weak edges.

5. View

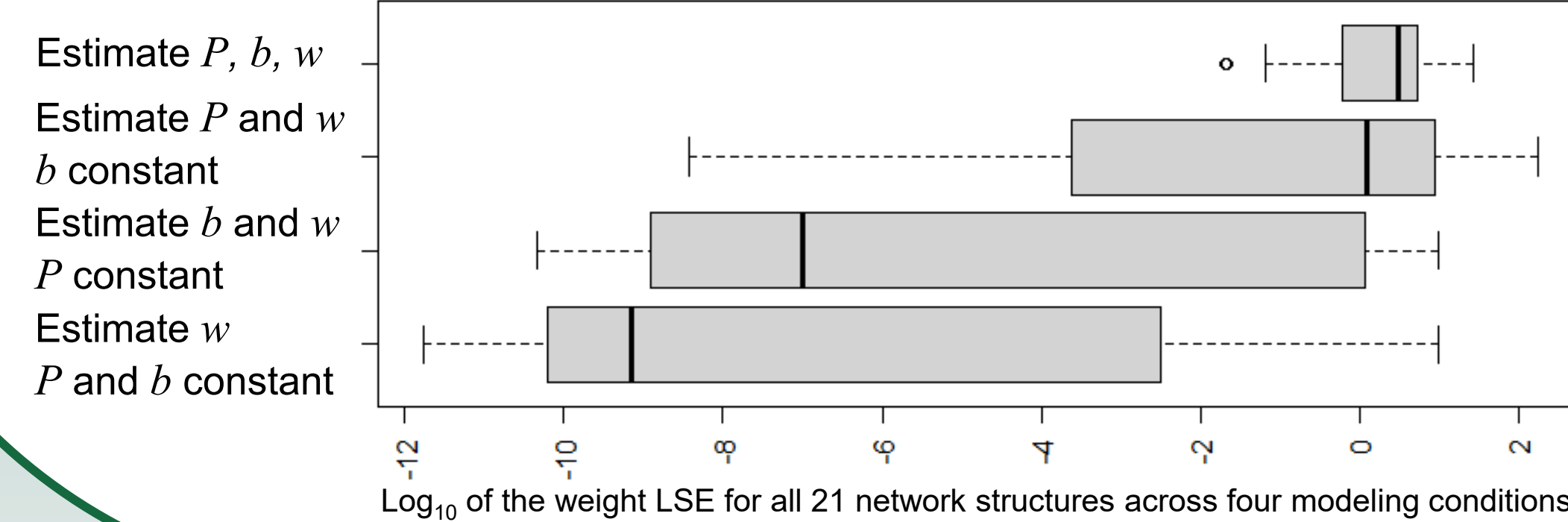
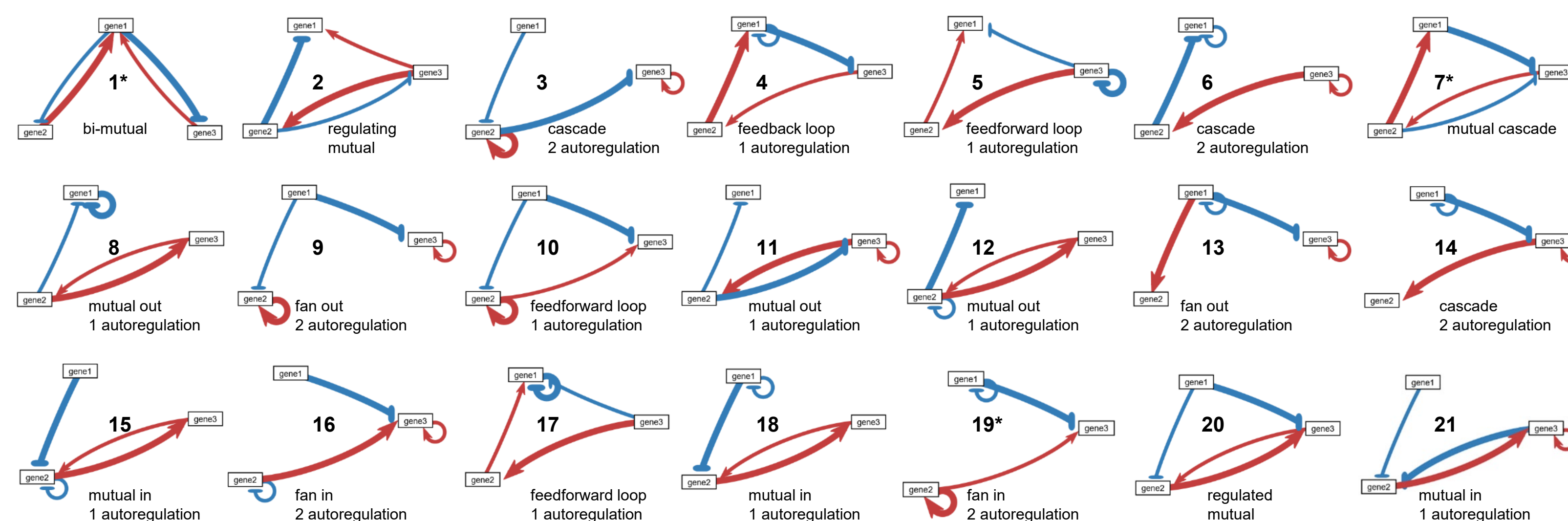
- Multiple viewport sizes available.
- Graph bounding box can be separated from viewport.
- Zooming and scrolling available.

6. Export

- Network data export to SIF or GraphML or full data to GRNmap-compatible Excel workbook, with yeast mRNA degradation rates from Neymotin et al. (2014).
- Image export to PNG, SVG, or PDF.



Model sensitivity was tested using all possible 3-gene, 4-edge “toy” networks. Simulated data was generated based on known parameters. Then the simulated data was used to estimate parameters to see if the model returned the known values.



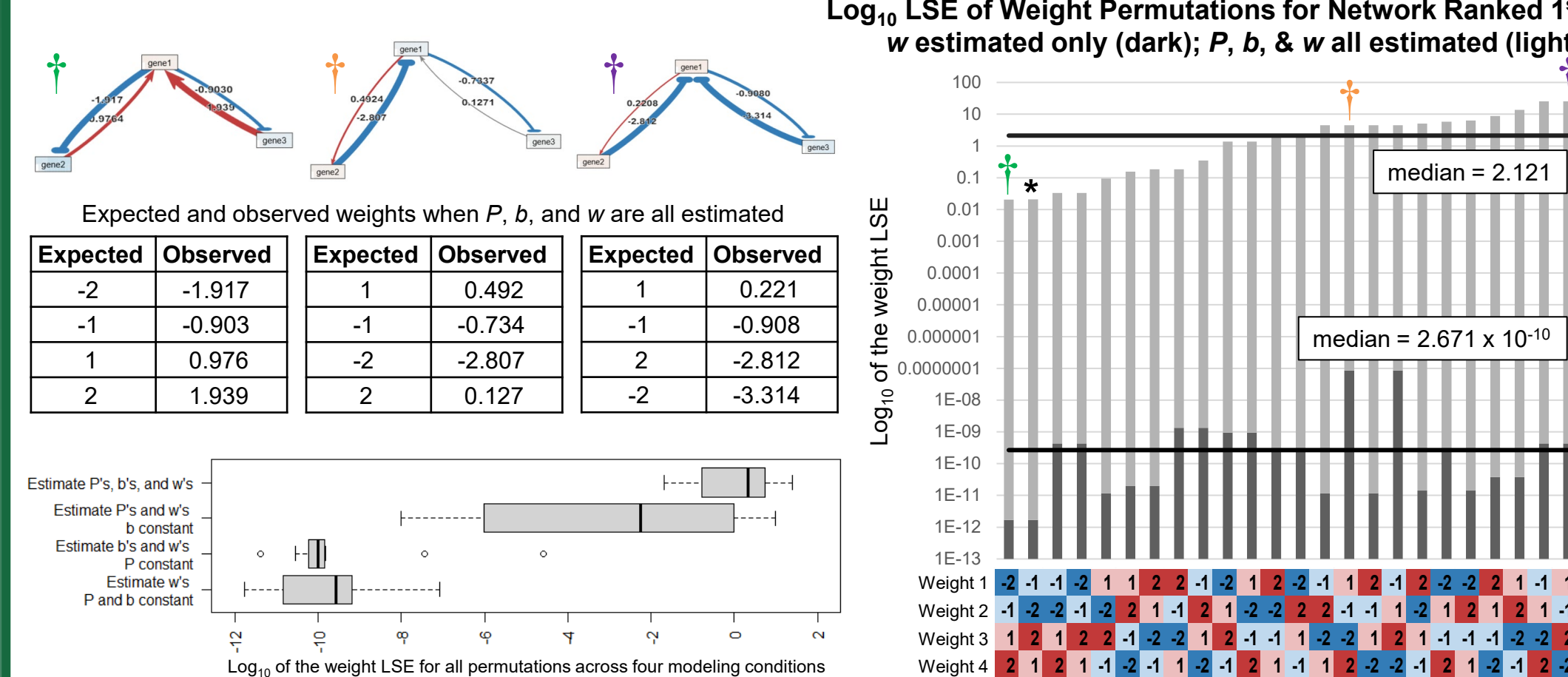
- Least squares error, LSE, was used to rank the performance of the networks relative to one another.**
- The error between the known weights and the estimated weights was computed as follows:
$$LSE = \sum_{i=1}^n (w_i - w'_i)^2$$
where w_i are the actual weights and w'_i are the estimated weights.
- Ranking based on the sum of the LSEs under the four conditions.

- When P and b were held constant, only the 4 weight parameters were estimated, compared to the 10 estimated when P , b , and w were all estimated.
- The performance of the model when P and b were held constant and when only P was held constant were relatively similar, suggesting that the issue is estimating the production rate, not necessarily the number of parameters estimated.

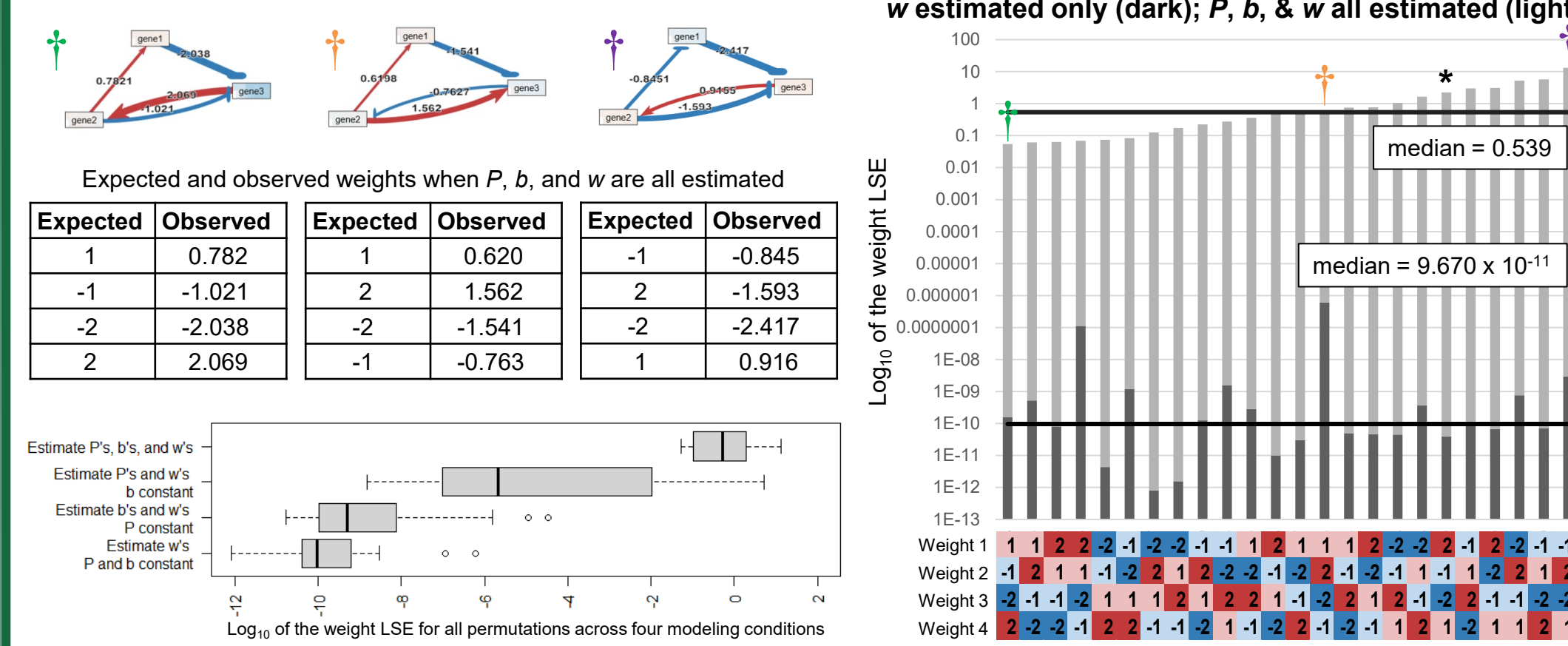
For the 3-gene, 4-edge “toy” networks, weight estimation precision is dependent on weight values

- Since weights were arbitrarily assigned (1, -1, 2, or -2), we also investigated if the rankings were solely based on network structure, or if they were also impacted by the weights assigned to the edges.
- All 24 permutations of the weight assignments to edges were modeled for the networks originally ranked 1st, 7th, and 19th to determine the impact of the weights.
- The LSE of the weights were calculated for each trial as before.

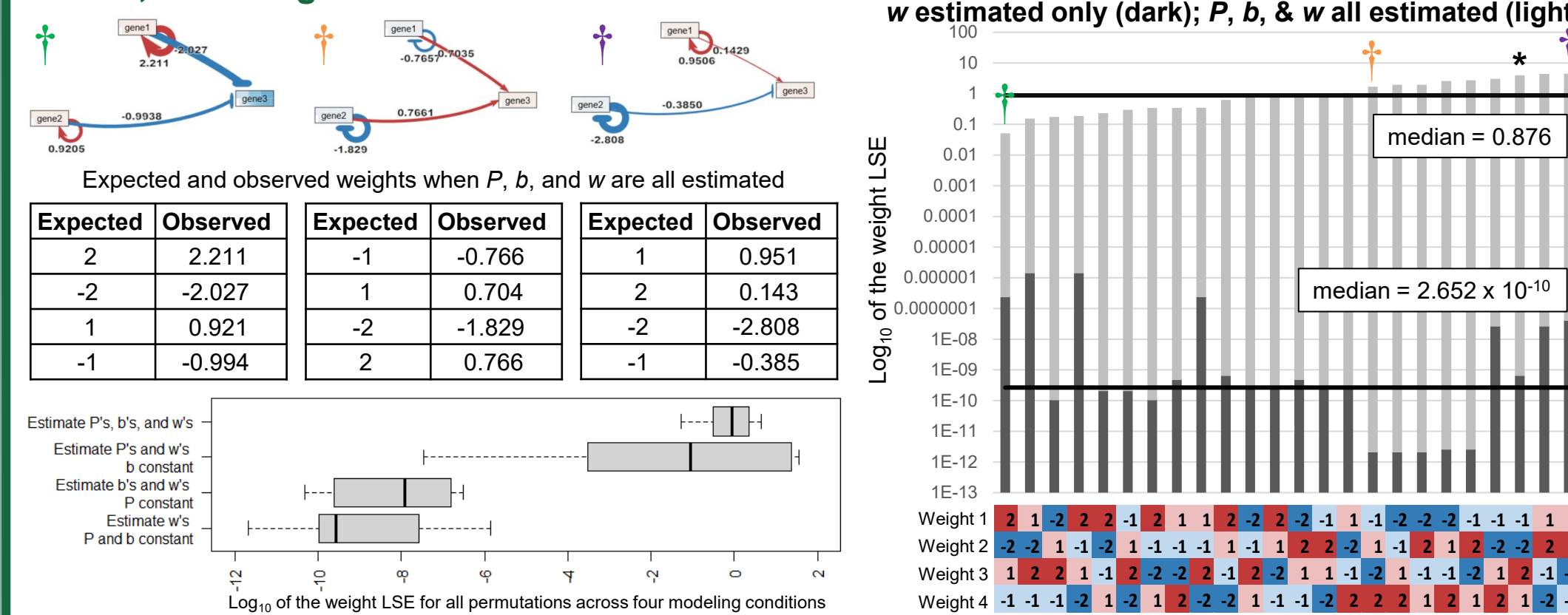
bi-mutual



mutual cascade



fan in, 2 autoregulation



- Generally, when only w is estimated, the LSE is very low (10^{-12} – 10^{-6}), so we ranked model performance when P , b , and w were all estimated, $LSE = 10^{-2}$ – 10^1 (see bar charts).
- The box and whisker plots show the same pattern that we saw previously where the median LSE is smaller in the two conditions where P is held constant.
- The mutual cascade family of networks performs better than the other two network families with smaller median LSEs for all estimation conditions.

Conclusions

- GRNmap and GRNsight are part of an integrated workflow to model and visualize GRNs in yeast.
- Investigation of 3-gene, 4-edge “toy” networks showed that performance of the model estimation is influenced by a combination of network structure and edge weights.

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Acknowledgments

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